# Python + Git/GitHub + Linux Roadmap (Aug–Dec 2025) 🐍💻

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## AUGUST – Git & GitHub Foundations

**Goal:** Understand version control and share code online.

**Topics:**

- **Git Basics**
  - Install Git
  - `git init`, `git add`, `git commit`
  - `git status`, `git log`
  - Branching (`git branch`, `git checkout`) and merging (`git merge`)
- **GitHub Basics**
  - Create GitHub account (with adult help)
  - Create a repository
  - Connect local repo to GitHub (`git remote add origin`)
  - Push (`git push`) and pull (`git pull`) changes

**Mini Projects:**

- Save and track changes for a Python "About Me" script.
- Create a repository for Python practice code.

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## SEPTEMBER – Linux Commands + Strings

**Goal:** Learn terminal basics and master string handling in Python.

**Linux Commands:**

- File & directory navigation: `pwd`, `ls`, `cd`
- File operations: `touch`, `mkdir`, `rm`, `rmdir`, `cp`, `mv`
- Viewing files: `cat`, `less`, `head`, `tail`
- Permissions: `chmod` basics
- System info: `whoami`, `uname -a`, `df -h`

**Python Strings:**

- Operations: concatenation, repetition, membership (`in`), slicing
- Traversing strings with loops
- Built-in methods: `len()`, `capitalize()`, `title()`, `lower()`, `upper()`, `count()`, `find()`, `index()`, `endswith()`, `startswith()`, `isalnum()`, `isalpha()`, `isdigit()`, `islower()`, `isupper()`, `isspace()`, `lstrip()`, `rstrip()`, `strip()`, `replace()`, `join()`, `partition()`, `split()`

**Mini Projects:**

- Count vowels in a string
- Find most frequent character in a string
- Check if a string is a palindrome

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## OCTOBER – Lists, Tuples & Dictionaries

**Goal:** Learn how to store, access, and manipulate collections of data.

**Lists:**

- Indexing, slicing, concatenation, repetition, membership
- Built-in methods: `len()`, `list()`, `append()`, `extend()`, `insert()`, `count()`, `index()`, `remove()`, `pop()`, `reverse()`, `sort()`, `sorted()`, `min()`, `max()`, `sum()`
- Nested lists
- Mini Projects: Find max, min, mean from a list; linear search; frequency counter

**Tuples:**

- Indexing, slicing, concatenation, repetition, membership
- Built-in methods: `len()`, `tuple()`, `count()`, `index()`, `sorted()`, `min()`, `max()`, `sum()`
- Nested tuples
- Mini Projects: Mean from tuple; search for number; count frequency of items

**Dictionaries:**

- Keys & values, adding/modifying items
- Built-in methods: `len()`, `dict()`, `keys()`, `values()`, `items()`, `get()`, `update()`, `del`, `clear()`, `fromkeys()`, `copy()`, `pop()`, `popitem()`, `setdefault()`, `max()`, `min()`, `sorted()`
- Mini Projects: Character count from string; employee salary storage & retrieval

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## NOVEMBER – OOP + Python Modules

**Goal:** Learn object-oriented programming and use standard Python libraries.

**OOP (Classes & Objects):**

- What is a class & object
- Defining a class with `__init__` method
- Attributes and methods
- Multiple objects from one class
- Example: Class for `Car` with attributes like brand, color, speed
- Mini Project: Student management system (store name, grade, marks)

**Modules:**

- Importing (`import`, `from ... import`)
- **math**: `pi`, `e`, `sqrt()`, `ceil()`, `floor()`, `pow()`, `fabs()`, `sin()`, `cos()`, `tan()`
- **random**: `random()`, `randint()`, `randrange()`
- **statistics**: `mean()`, `median()`, `mode()`

- Mini Project: Math quiz generator, random password generator, student marks statistics calculator

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### DECEMBER – Integration Projects + Streamlit

**Goal:** Combine all skills into real-world projects and share online.

**Streamlit Basics:**

- Install (`pip install streamlit`)
- Create & run first app (`streamlit run app.py`)
- Widgets: text input, sliders, buttons
- Display text, images, and charts
- Deploy to Streamlit Cloud

**Final Projects:**

- Quiz game web app
- Student report card generator (uses classes, lists, dictionaries, statistics)